

A Colexification Network: The Structure of Human Language and Thought

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1. Introduction

If you speak English, you use the word *leg* for the entire lower limb. If you speak Finnish, you use two different words: one for the thigh and one for the lower leg. If you speak Turkish, a single word covers both the foot and the leg together. These differences are not accidents—they reflect genuine variation in how different cultures carve up the same physical reality into distinct concepts. But they also raise a deeper question: beneath all this variation, is there a universal structure to how humans organize meaning? Are there concepts that every language, regardless of geography or ancestry, tends to group together?

This paper approaches that question through the lens of network science. We analyze *colexification*: the phenomenon where a single word in a language covers multiple concepts simultaneously. When a language uses one word for both *hand* and *arm*, it is not merely an accident of vocabulary; it reflects a cognitive judgment that these two concepts belong together closely enough to share a name. When dozens of unrelated language families independently make the same judgment, that overlap becomes evidence of something universal about human thought.

Using the CLICS database (Rzyski et al., 2020), which catalogs colexifications across hundreds of languages and language families, we construct a network in which concepts are nodes and edges connect concepts that are colexified in at least three unrelated language families. The result is a map of human semantic space, not as any single language constructs it, but as the aggregate of linguistic behavior across cultures suggests it universally is.

Network science is well suited to this question. Prior work has applied graph-theoretic methods to phonological structure (Shoemark et al., 2016), syntactic organization (Ferrer i Cancho et al., 2004), and semantic association (Steyvers & Tenenbaum, 2005), finding that language networks tend to exhibit small-world properties and scale-free degree distributions. These structural regularities suggest that the organization of language is

not arbitrary, it reflects underlying cognitive and communicative pressures. Our contribution is to ask not just whether the colexification network has these properties, but what its specific structure reveals about the universal architecture of human meaning, and in particular whether concepts of different grammatical categories—verbs, nouns, and adjectives—play structurally distinct roles within that space.

Our analysis produces four main findings. First, the colexification network has strong community structure (modularity $Q = 0.82$), with 32 distinct semantic neighborhoods whose membership is interpretable and cross-linguistically stable. Second, this structure sharpens as we demand greater cross-linguistic agreement: modularity increases from 0.82 to 0.92 as we raise the threshold for what counts as a valid colexification edge, before collapsing when only 13 concepts remain. Those 13 concepts (the most universal in the entire database) are exclusively about the surface and substance of living bodies: skin, fur, flesh, hair, feather, bark.

Third, centrality analysis reveals a structural division of labor by part of speech: verbs score significantly higher than chance on betweenness and closeness centrality ($z = 3.44$ and $z = 8.85$ respectively), while nouns score significantly higher on eigenvector centrality ($z = 2.46$). Verbs are the connective tissue of the network, bridging communities and broadcasting across semantic space—while nouns form its dense local hubs.

Fourth, and most surprisingly, this pattern reverses at maximum universality: among the most cross-linguistically stable concepts, nouns dominate and verbs disappear entirely, suggesting that the core of human thought is anchored in concrete physical things rather than actions.

Together these findings suggest that the semantic space encoded across human languages has a layered structure: a universal foundation of concrete nouns anchored in embodied physical experience, bridged at the cultural level by a widespread but not universal skeleton of action verbs. We discuss implications for theories

of semantic universals and the relationship between vocabulary structure and cognitive organization.

2. Background

The premise that language can be modeled as a complex network has yielded significant insights across both syntax and semantics. To contextualize our investigation into the structural role of grammatical categories, we build upon three foundational applications of network science to linguistic structure: one examining the cognitive boundaries of meaning across cultures, one examining the general topology of semantic association, and one examining the physical architecture of grammar.

The methodological foundation for this project relies on Jackson et al. (2019), who demonstrated that colexification networks can reliably map the cognitive structure of human semantics. Utilizing the CLICS database, Jackson and colleagues constructed networks of emotion concepts across thousands of languages, revealing distinct community structures within emotion vocabulary. Their analysis showed that while the specific boundaries of these neighborhoods vary with geographic and cultural proximity, the underlying mechanism of colexification is a reliable proxy for cognitive categorization — languages that share more history and environment tend to carve up emotional meaning more similarly.

The quality and impact of the Jackson et al. study are substantial; it successfully bridges computational linguistics and anthropology. But its value to our project lies also in its limitations. By restricting their scope to a single, highly abstract domain and focusing primarily on cultural variance, their analysis leaves the broader structure of human semantic space unexplored. Our project adopts their colexification methodology but scales it to the entirety of the CLICS database. Furthermore, rather than examining how cultures differ, we use thresholding to isolate what is strictly universal, shifting the focus from cultural variation to the underlying structure of meaning.

A additional perspective comes from Steyvers and Tenenbaum (2005), who analyzed the large-scale structure of three semantic networks built from word association norms, WordNet, and Roget’s Thesaurus. Across all three, they found consistent small-world properties and scale-free degree distributions, and argued that these properties emerge from the same growth pro-

cess: new concepts are added to the lexicon preferentially near already well-connected concepts, producing hubs and clusters naturally over time. Their work establishes that semantic networks in general (not just colexification networks specifically) have non-random, structured topologies that reflect cognitive organization. This provides an important baseline against which our results can be compared: the community structure and centrality patterns we observe are not merely properties of the CLICS dataset, but consistent with broader findings about how semantic memory is organized.

This background cites the colexification methodology to Jackson et al. as well as the general topology of semantic networks to Steyvers and Tenenbaum. Ferrer i Cancho et al. (2004) then establishes the laws that govern how words interact in grammar. In their analysis of syntactic dependency networks, they constructed graphs based on how words link together structurally in sentences, finding that human grammar forms a small-world network characterized by highly clustered local neighborhoods and short global path lengths. They argued this topology is not accidental but reflects cognitive and evolutionary pressure toward efficient communication: the network requires highly connected hub words to allow rapid traversal of syntactic space, a property observed across all languages studied.

Ferrer i Cancho’s analysis was confined to syntax, the arrangement of words in sentences. Our analysis bridges this literature with the semantic work of Jackson and Steyvers and Tenenbaum. If syntactic networks are optimized for efficient communication via structural hubs, and colexification networks reflect how humans categorize reality across cultures, then grammatical categories themselves may serve distinct and measurable roles within the semantic network. By mapping part-of-speech data onto a universal colexification network and testing for statistically significant differences in centrality by grammatical class, this project asks whether the abstract distinctions of grammar are visible in the structure of meaning.

3. Methods

3.1. Data and Network Construction

The primary dataset for this project is the CLICS4 database (Rzymyski et al., 2020), a curated repository of colexification data compiled from hundreds of linguistic sources spanning over 3,000 concepts and 250 language families. Each entry in the database records a

pair of concepts that a given language expresses with a single word, along with the language family to which that language belongs. From this, we constructed an undirected weighted network in which nodes represent semantic concepts and edges connect concept pairs that are colexified in at least three distinct language families. The family-count threshold was chosen deliberately: requiring agreement across unrelated families filters out accidental homonyms and individual vocabulary choices, leaving only connections that reflect recurring cross-cultural cognitive judgments. Edge weights were set equal to the number of language families sharing each colexification. Network construction and all subsequent analysis were carried out in Python using NetworkX, with community detection performed via the Louvain algorithm as implemented in the `python-louvain` library.

The resulting network contained 1,386 nodes, of which 1,300 belonged to a single giant connected component: 93.8% of the full network. All following analysis was conducted on this giant component. Basic topology metrics confirmed small-world properties consistent with prior semantic network studies: average path length of 4.89 steps and a clustering coefficient of 0.30. Louvain community detection yielded 32 communities with a modularity score of $Q = 0.82$, well above what random graph models would predict.

3.2. Universality Gradient and Threshold Analysis

To investigate the relationship between cross-linguistic universality and network structure, we constructed four versions of the network at progressively stricter thresholds: ≥ 3 , ≥ 5 , ≥ 10 , and ≥ 20 language families. At each threshold we recomputed the giant component size, re-ran the Louvain algorithm, and recorded the resulting modularity score. This analysis was originally conceived as a robustness check, a way to confirm that the community structure observed at the base threshold was not an artifact of noisy or peripheral data. What we found instead became central to the project’s thesis. As the threshold increased from 3 to 10 families, modularity rose consistently from 0.82 to 0.92, indicating that stricter universality constraints produce sharper and more coherent semantic neighborhoods. At the maximum threshold of 20 families, the network collapsed to just 13 nodes — every one of them a concrete anatomical noun related to the surface of a living body. This was not the expected result. The original intention had been to use the high-threshold network as the founda-

tion for an extension involving Universal Dependencies syntactic (grammatical) treebanks, comparing semantic and syntactic proximity across the same concept set. The collapse of the network to 13 nodes changed the direction of the project. Our thesis incorporates the finding that the most universal concepts in human language are not abstract, relational, or verbal, but are physical nouns grounded in the body. We refer to this as the “hourglass” structure of human thought.

3.3. Part-of-Speech Tagging

To investigate whether grammatical category predicts structural role in the network, each concept node was assigned a part-of-speech label: verb, noun, or adjective. Our first approach used the NLTK WordNet interface, a large lexical database that organizes English words into sets of synonyms (called *synsets*), each tagged with a grammatical category. We assigned each concept the category of its most common entry in the database. This failed immediately: WordNet defaults to the most frequent dictionary sense of a word, which are disproportionately nouns in English. Core concepts such as SAY, DIE, DO, and COME were all labeled as nouns, collapsing the verb category and making any centrality comparison between parts of speech meaningless.

We then revised the tagger to vote across all available dictionary entries for a given word rather than defaulting to the most common one — if a word appeared more often as a verb than a noun across all its definitions, it was labeled a verb. This improved accuracy considerably but introduced a new problem: compound CLICS concept labels such as “YOUNGER SISTER (OF MAN)” and “YOUNGER BROTHER (OF WOMAN)” were being parsed as adjective phrases due to the leading modifier word, mislabeling the entire kinship community as adjectives.

The final solution was a hybrid tagger combining the voting heuristic with a manual override dictionary covering approximately 40 high-frequency or structurally important concepts — core verbs, kinship terms, and frequently misclassified compound phrases. After applying this fix, the network composition settled at 787 nouns, 393 verbs, and 120 adjectives, and the kinship community was correctly identified as nouns. The statistical validation of these labels is described in the following subsection; the corrected assignments produced the centrality z-scores reported in the Results section.

3.4. Statistical Validation

To test whether the observed centrality differences by part of speech were statistically significant rather than a consequence of class size differences, we implemented a Monte Carlo permutation test for each centrality measure. For a given part-of-speech class and centrality metric, we computed the actual mean centrality score across all nodes in that class, then generated a null distribution by performing 1,000 random shuffles of the centrality scores across all nodes and recording the mean of a randomly drawn subset of the same size. The z-score of the observed mean against this null distribution provides a measure of how many standard deviations the actual result lies from what chance would predict.

An early implementation of this test failed with a `ValueError: array is read-only` exception, caused by Pandas protecting the internal memory of its DataFrame columns against in-place modification. The fix was to extract centrality scores into independent NumPy arrays before shuffling, decoupling the permutation procedure from the DataFrame entirely. The corrected implementation ran 1,000 permutations per class per metric without issue.

3.5. Visualization

Several visualization approaches were attempted before settling on the figures presented in the Results section. Initial attempts to compare centrality distributions across parts of speech using standard histograms were unreadable in practice: the severe class imbalance between nouns (787 nodes) and verbs (393 nodes) meant that raw frequency bars for nouns visually overwhelmed those for verbs, obscuring the structural differences we were trying to show. We switched to Seaborn kernel density estimates with `common_norm=False`, which normalizes each class independently to a common scale. This made the distributional differences (particularly the heavy tail of high-betweenness verbs) immediately visible. Community structure was visualized using both NetworkX spring layouts for subgraph analysis and Gephi’s ForceAtlas2 algorithm for the full network, with nodes sized by closeness centrality and colored by part of speech.

4. Results

4.1. The Global Structure of Human Semantic Space

The colexification network constructed from the CLICS4 database contains 1,300 concepts in its giant connected component, retaining 93.8% of all nodes. This is a sign that human semantic space is largely continuous rather than fragmented into islands of meaning. The network exhibits the hallmarks of a small-world graph: an average path length of 4.89 steps and a clustering coefficient of 0.30. Any two concepts in the network are reachable from one another in under five steps on average, yet concepts cluster into tight local neighborhoods far more than a random graph of the same size would produce. This combination (local density, global reachability) is a sign the system is under simultaneous pressure to organize meaning into coherent categories while keeping those categories accessible to one another.

Louvain community detection identifies 33 distinct neighborhoods with a modularity score of $Q = 0.82$. This is a high value: random graphs with equivalent degree sequences produce modularity scores near zero, and values above 0.3 are generally considered significant. The communities themselves are semantically very interpretable. Neighborhood 1 (106 concepts) groups actions of force and violence — DIE, KILL, SHOOT, DIG, THROW. Neighborhood 2 (104 concepts) groups perception, cognition, and speech — SAY, SEE, HEAR, KNOW, THINK. Neighborhood 3 (85 concepts) groups motion and biological life — WALK, COME, GO, BE ALIVE, GROW. Neighborhood 7 (61 concepts) groups celestial and meteorological phenomena — SKY, SUN, CLOUD, WEATHER, HEAVEN. These groupings come from patterns across thousands of independent linguistic decisions made across unrelated cultures. The way they resolve into recognizable domains of human experience suggests that the community structure of the network is a genuine feature of cognition instead of an artifact in the data. Figure 1 shows the full network visualized with nodes colored by community, nodes and labels sized by pagerank and betweenness centrality respectively.

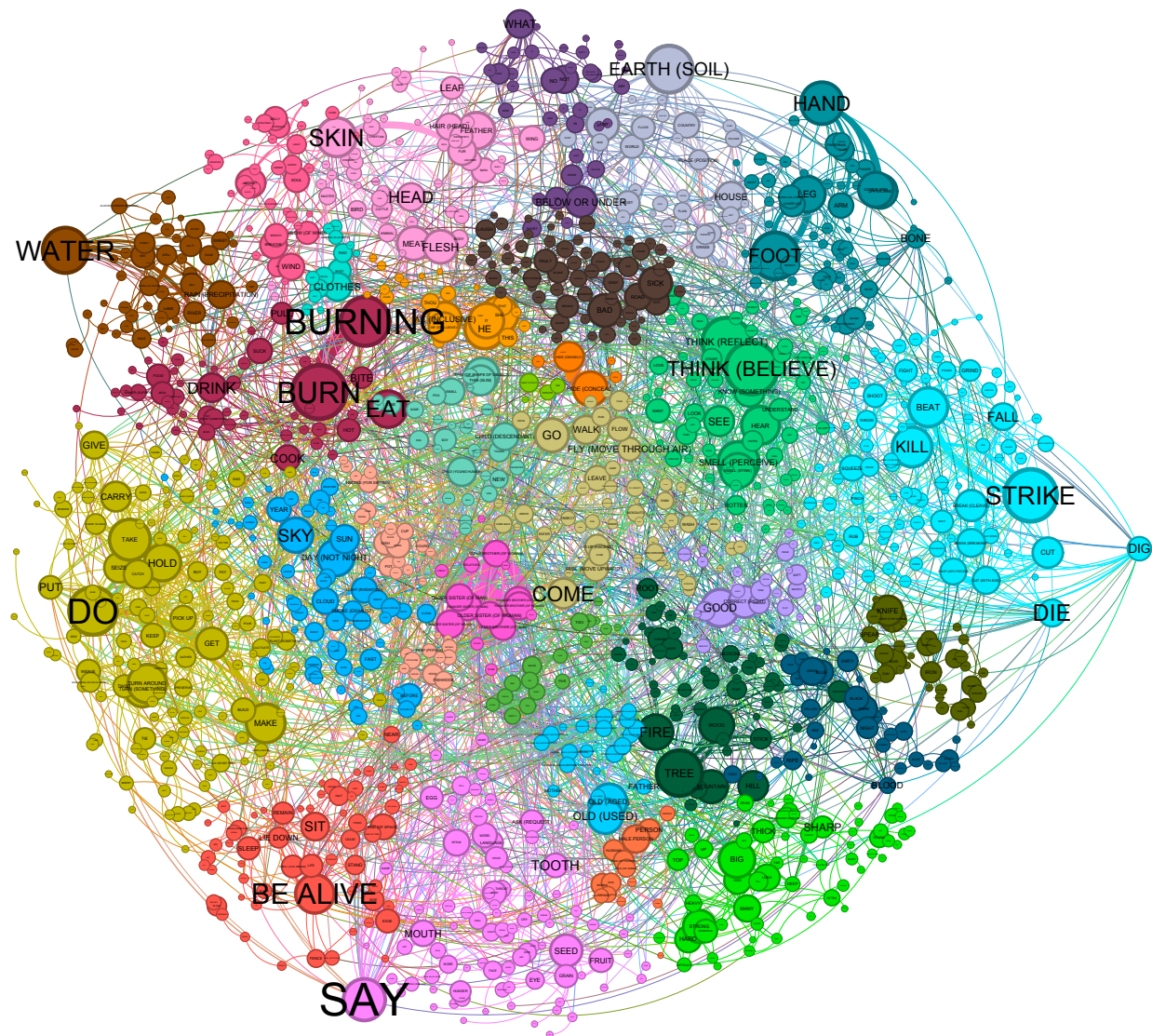


Figure 1: The full collexification network (1,300 concepts), with nodes colored by community and sized by PageRank. Labeled nodes have betweenness centrality above 0.015. Identifiable communities include Violence/Force (light blue), Perception/Speech (green), Motion/Life (red), and Sky/Celestial (dark blue).

4.2. The Universality Gradient and the Hourglass Structure

The base network, constructed at a threshold of ≥ 3 language families, includes any colexification shared by at least three unrelated language families. Raising this threshold progressively filters out connections that, while widespread, fall short of true universality. Table 1 summarizes the effect of this filtering on network size and community structure.

Threshold	Concepts	Communities	Modularity
≥ 3 families	1,300	33	0.82
≥ 5 families	1,001	34	0.89
≥ 10 families	335	25	0.92
≥ 20 families	13	3	0.54

Table 1: Network properties at increasing universality thresholds.

There are two main findings in this table. First, modularity increases as the threshold rises from 3 to 10 families (from 0.82 to 0.92). Removing nodes from a network can disrupt its community structure just as easily as it can strengthen it. The fact that modularity rises suggests that the edges removed at higher thresholds are predominantly the ambiguous, cross-community connections (the weak bridges between neighborhoods) while the edges retained are the stronger, within-community ones. In other words, the most universal colexifications are also the most semantically coherent ones. Universal patterns in Human thought are more tightly organized than culturally specific ones.

Second, at the maximum threshold of ≥ 20 fami-

lies, the network collapses to just 13 concepts. These are listed in full in Figure 3. Every single one belongs to a single category: the surface and substance of living bodies. SKIN, FUR, FEATHER, BARK, and LEATHER are coverings. HAIR, HAIR (HEAD), and HAIR (BODY) are filaments that grow from surfaces. FLESH, MEAT, and BODY are the substance beneath. WING is the biological extension. Across every language family in the database, including those with no shared ancestry or geographic proximity, the same small set of concepts about the biological boundary between a living thing and the world are grouped together. This is the deepest layer of the network: not abstract, social, or divine, but physical. We refer to this structure as the *hourglass*: broad and richly connected at the cultural level, narrowing to a universal core of physical experience at the base.

4.3. Centrality and the Division of Labor by Part of Speech

The most structurally important concepts in the network, as measured by betweenness, closeness, degree, and eigenvector centrality, are not distributed randomly across grammatical categories. Table 2 reports the top five concepts for each centrality measure. Betweenness, closeness, and degree are dominated entirely by verbs: SAY, DIE, DO, EAT, COME, BURNING. These are concepts of action and existence, what it means to be alive in the world. Eigenvector centrality stands out as the only measure where its top five are all kinship terms, and all nouns. To understand why this contrast is meaningful, it is necessary to distinguish what each measure captures.

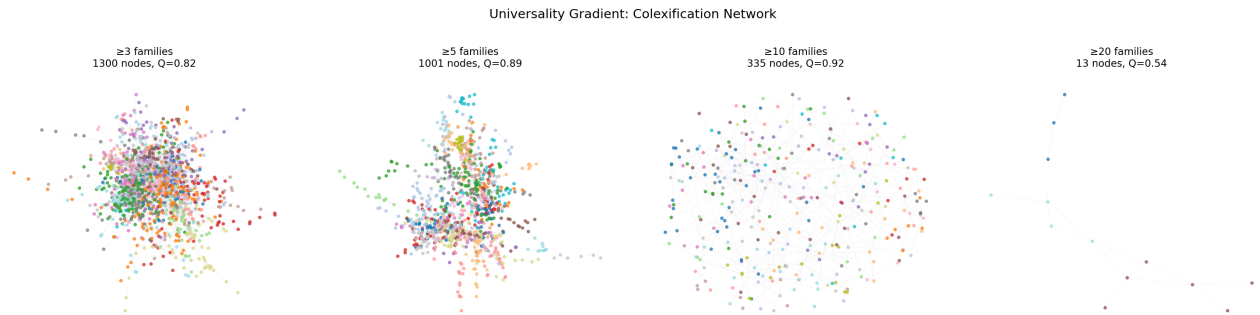


Figure 2: The universality gradient across four thresholds. Each panel shows the giant component colored by community, shrinking as the minimum language family threshold increases from ≥ 3 (1,300 nodes, $Q = 0.82$) to ≥ 20 (13 nodes, $Q = 0.54$). Community structure sharpens visibly as the threshold rises before collapsing to a core of 13 anatomical nouns.

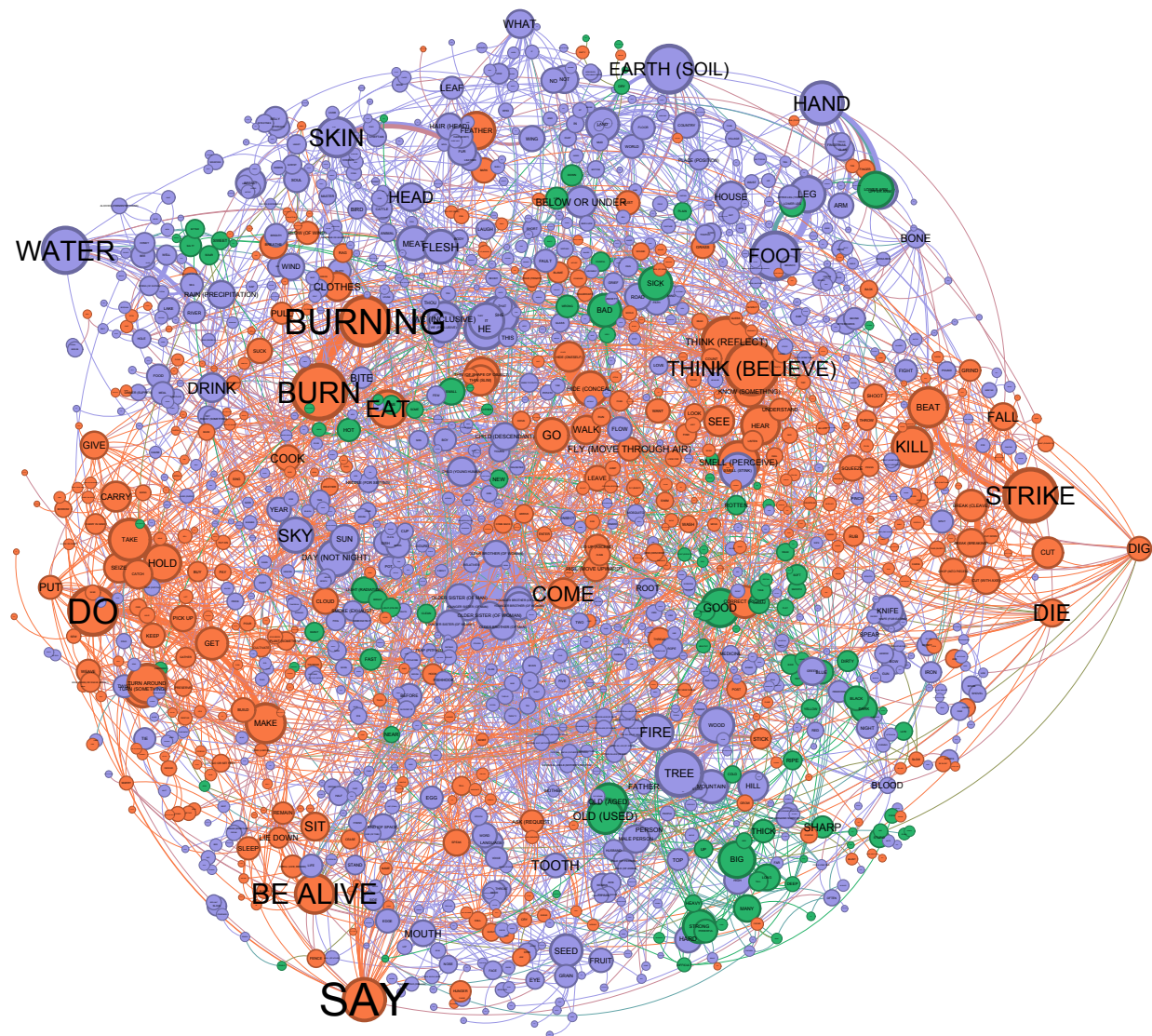


Figure 3: The full colexification network colored by part of speech: verbs (coral), nouns (purple), adjectives (teal). Node size reflects PageRank; labeled nodes have highest betweenness centrality. Verbs concentrate toward the structural center of the network while nouns cluster toward the periphery.

Measure	Top Concepts	Score
Betweenness	SAY	0.082
	DIE	0.062
	BE ALIVE	0.053
	DO	0.052
	BURNING	0.052
Closeness	DIE	0.301
	EAT	0.299
	BURNING	0.299
	COME	0.298
	SAY	0.298
Degree	SAY	0.030
	DO	0.027
	EAT	0.021
	STRIKE	0.021
	DIE	0.020
Eigenvector	YOUNGER BROTHER (OF WOMAN)	0.366
	YOUNGER BROTHER (OF MAN)	0.362
	YOUNGER SISTER (OF MAN)	0.361
	YOUNGER SISTER (OF WOMAN)	0.361
	OLDER SISTER (OF MAN)	0.342
PageRank	BURN (SOMETHING)	0.0035
	STRIKE	0.0034
	THINK (BELIEVE)	0.0034
	THINK (REFLECT)	0.0032
	BURNING	0.0032

Table 2: Top five concepts by centrality measure.

Betweenness centrality identifies concepts that sit on the shortest paths between many other concepts (the bridges). Closeness centrality identifies concepts that can reach all other concepts quickly. Eigenvector centrality, by contrast, identifies concepts whose neighbors are themselves highly connected, so these five nodes are embedded in the richest, most mutually reinforcing local neighborhoods.

These familial terms are not globally central, they sit inside an exceptionally dense cluster of meaning where every concept is closely connected to every other. Kinship vocabulary, it appears, forms the most elaborated and internally consistent semantic neighborhood in the entire network.

PageRank differs from eigenvector because it normalizes each neighbor’s contribution by degree. So kinship terms scores lower than a community that is both internally dense and outward facing.

To test whether this grammatical pattern is statistically significant rather than a coincidence of the top-five rankings, we ran a Monte Carlo permutation test for each centrality measure and each part-of-speech class, generating a null distribution from 1,000 random shuffles of centrality scores across nodes. Table 3 reports the resulting z-scores.

POS	Betweenness	Eigenvector	Closeness
Verb	+3.45	−1.99	+8.00
Noun	−3.87	+2.44	−9.41
Adjective	+1.06	−0.93	+2.23

Table 3: Z-scores from Monte Carlo permutation tests (1,000 shuffles). Values above $|2|$ are statistically significant.

Agreeing with the centrality measures, verbs score 3.45 standard deviations above chance on betweenness and 8.00 standard deviations above chance on closeness. Nouns score 2.44 standard deviations above chance on eigenvector, while scoring significantly *below* chance on betweenness ($z = -3.87$) and closeness ($z = -9.41$). Adjectives show no significant pattern on any measure. Verbs are the highways of this semantic network: they connect distant communities and enable fast traversal in both speech and thought. Nouns are the neighborhoods: dense, locally rich, but globally irrelevant. Figure 4 shows the full centrality distributions by part of speech, with the verb fat-tail in betweenness and closeness clearly visible.

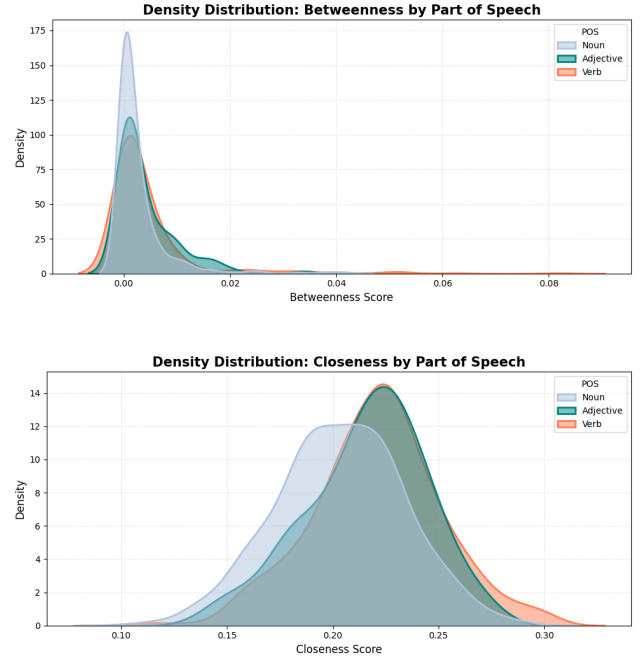
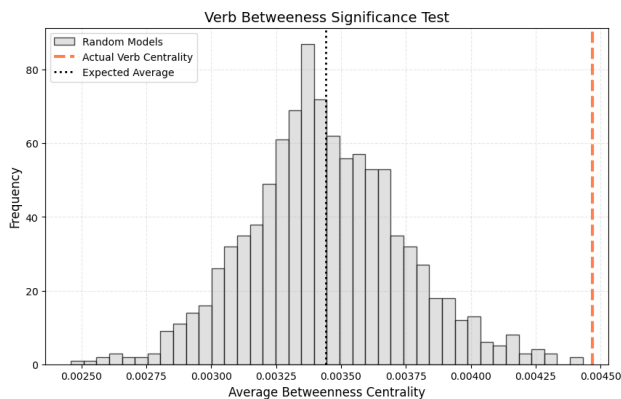


Figure 4: Kernel density estimates of betweenness (top) and closeness (bottom) centrality by part of speech, normalized independently (`common_norm=False`). The heavy right tail of the verb distribution in both panels reflects the disproportionate concentration of verbs among the network’s most central nodes.



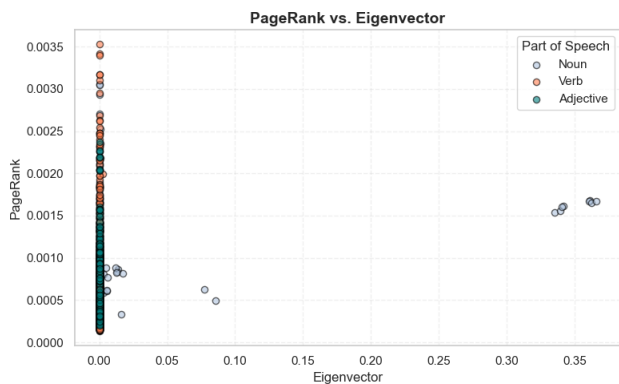
(a) Null distribution of average verb betweenness centrality from 1,000 random permutations (gray), with the observed value marked (dashed line, $Z = 3.45$). The observed verb betweenness falls 3.45 standard deviations above the null mean, rejecting the hypothesis that verbs are structurally central by chance alone.

4.4. The Universality Reversal

The structural dominance of verbs, however, is not universal. Table 1 showed that as the threshold rises, the network contracts. Figure 3 shows what happens to verb centrality as it does. At ≥ 3 families, the verb betweenness z-score is $+3.27$: verbs are strongly over-represented among the network’s bridges. At ≥ 5 families the signal weakens to $+2.57$ but remains significant. At ≥ 10 families it drops to -1.19 : verbs are no longer special. At ≥ 20 families it is -0.16 : essentially zero, and in any case only 2 of the 13 surviving concepts are verbs at all.

Here is the so called hourglass structure of this semantic space. At the cultural level, colexifications shared by a handful of language families, verbs dominate the structural framework. At the universal level, colexifications shared across the most unrelated language families on earth, verbs disappear and concrete nouns take over entirely. The most cross-linguistically stable semantic groupings are not about what humans *do*, but about what they *are made of*.

This finding has an intuitive explanation. Verbs encode actions and events, which are necessarily embedded in cultural context: what counts as a single act of “giving” or “striking” varies with social structure, technology, and environment. The verb-dominated communities we observe: violence, agency, motion, perception, are universally present but culturally variable in their internal organization. Concrete physical nouns, by contrast, are grounded in perceptual reality that does



(b) Betweenness vs. eigenvector centrality for all 1,300 concepts, colored by part of speech. Verbs (coral) cluster toward high betweenness and low eigenvector; nouns (steel blue) toward high eigenvector and low betweenness, reflecting their distinct structural roles as bridges and hubs respectively.

not vary across cultures. Every human body has skin, hair, and flesh. Every living organism has a surface that separates it from the world. These concepts do not need a cultural foundation to be grouped together.

4.5. The Reach of SAY: A Case Study in Semantic Bridging

The concept SAY occupies an unusual position in the network. It is the highest-betweenness node in the entire graph (0.082), the highest-degree node (0.030), and the fifth-highest closeness node (0.298). By every measure that captures global connectivity, SAY is the most structurally important concept in human semantic space. Its 39 direct neighbors span 8 distinct semantic communities, including domains as distant as EAT, FLY (verb), DIRTY, GRIND, CLOAK, PICK UP, and DAWN. This is not a concept embedded in a single neighborhood; it touches nearly every other neighborhood.

Why should SAY be so central? One interpretation is functional: speech is the mechanism through which humans navigate and transmit all other categories of experience. To speak is to act on perception, to express bodily states, to perform social roles, to describe the physical world. A concept that encodes the act of linguistic expression may naturally colexify with concepts from every domain simply because language *about* everything else is expressed through the same basic act of saying.

The Ego Network of 'SAY'

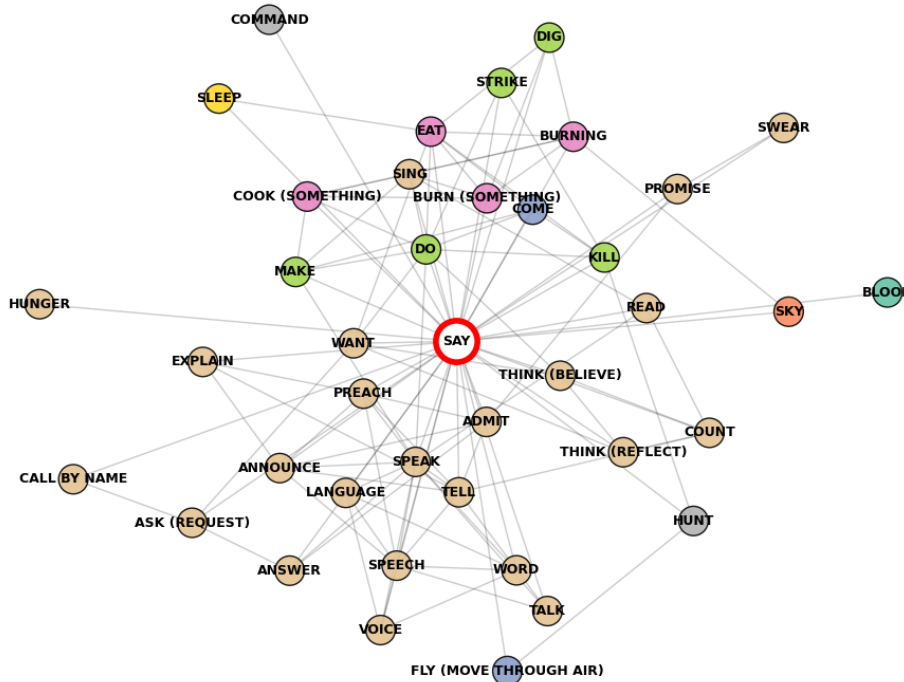


Figure 6: One-hop ego network of SAY (center), showing all 39 direct neighbors colored by community membership. SAY connects concepts from 8 distinct semantic communities simultaneously, including domains as distant as EAT and FLY.

The structural centrality of SAY in the network may be a result of the fact that language is the medium through which all other semantic categories become communicable. Figure 7 shows the ego network of SAY with neighbors colored by community.

4.6. Community-Level Structure and the Verb-Noun Divide

The grammatical division of labor is visible not only at the level of individual concepts but at the level of entire communities. Table 4 summarizes the POS composition of the ten largest neighborhoods.

The pattern is consistent and, communities organized around *what humans do* (violence, motion, perception, acquisition) are between 61% and 78% verbal. Communities organized around *what exists in the world* (body parts, space, sky, water, nature) are between 7% and 22% verbal. This is a structural divide. The network is organized into action-dominated and entity-dominated regions, and the boundary between them maps cleanly

Neighborhood	Label	Nouns	Verbs	% Verb
1	Violence / Force	34	70	66%
2	Perception / Speech	36	63	61%
3	Motion / Life	18	66	78%
4	Face / Body Parts	67	7	9%
5	Agency / Acquisition	25	46	64%
6	Space / Scale	37	5	7%
7	Sky / Celestial	47	8	13%
8	Color / Darkness	31	13	22%
9	Water / Landscape	40	7	13%
10	Nature / Flora	44	8	15%

Table 4: POS composition of the ten largest semantic communities.

onto the grammatical distinction between verbs and nouns. Figure 6 shows the full community-level POS breakdown across all 33 neighborhoods.

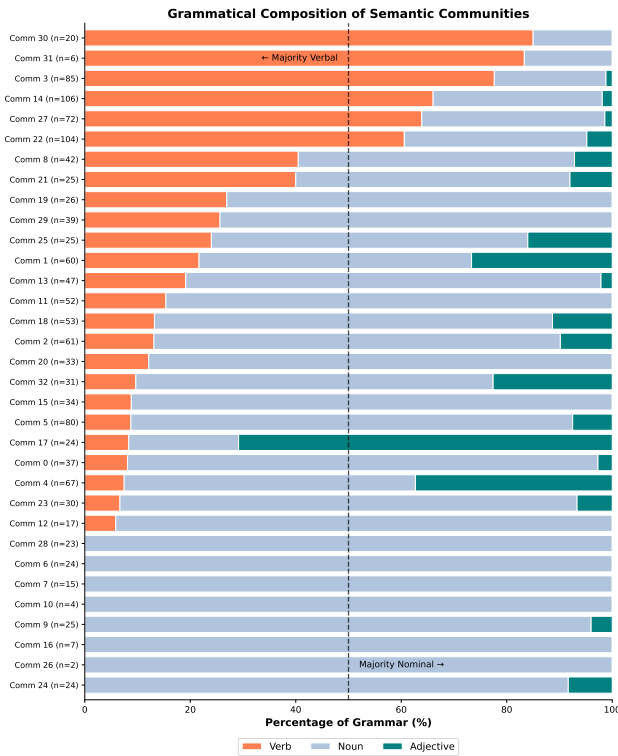


Figure 7: Part-of-speech composition of all 33 semantic communities, sorted by verb percentage. Action-oriented communities (top) are majority-verbal; entity-oriented communities (bottom) are majority-nominal. The boundary between the two regions maps onto the grammatical distinction between verbs and nouns.

This finding connects the network and community level results, verbs function as both global bridges and as local majorities in action domains. Verbs dominate more than the centrality scores, they are also concentrated in the communities that themselves serve as bridges between the more noun-heavy, entity-based neighborhoods. The Violence/Force community (Neighborhood 1) and the Motion/Life community (Neighborhood 3) are both majority-verb and both contain several of the network’s top betweenness nodes. The structural importance of verbs at the global level is in part a result of their concentration in these major communities.

5. Conclusion

This paper asked whether the structure of human semantic space, as revealed by a cross-linguistic colexification network, has universal properties that network science can detect and characterize. The answer is yes, and the structure is more specific than would be expected.

The colexification network is not just a small-world graph with modular communities, a result that prior work on semantic networks would have predicted. It has a layered structure with clear internal logic. At the broadest level, the network organizes human experience into 33 interpretable semantic neighborhoods, with action-oriented communities (violence, motion, perception, agency) playing a distinct role from entity-oriented ones (body, sky, water, nature). This division is not imposed by the analysis, it emerges from the data and is confirmed statistically by part-of-speech z-scores. Verbs are the bridges in this space: connecting communities and broadcasting meaning across the network. Nouns are the neighborhoods: dense, locally rich, but globally peripheral.

What we did not anticipate was that this pattern reverses completely under maximum universality constraints. The hourglass finding, that the 13 most cross-linguistically stable concepts are all concrete anatomical nouns, reframes the verb result rather than contradicting it. Verbs are culturally widespread bridges, while nouns are the universal foundation. The skeleton of meaning that every human language converges on is not about action or social structure. It is about the surface of a living body. This is perhaps the most unexpected result of the project, and the one most worth pursuing further.

Several natural extensions present themselves. The most immediate would be to investigate whether the hourglass structure holds within individual language families, does the same noun-core emerge when the analysis is restricted to, say, Austronesian or Niger-Congo languages, or does the universal core only appear when unrelated families are combined? A second extension would address the syntax question that the collapse of the high-threshold network made unworkable here: using the Universal Dependencies treebanks to test whether concepts that are semantically adjacent in the colexification network also tend to appear syntactically close in sentences across the same languages. If the verb-dominated communities that serve as semantic bridges also contain the most frequent syntactic root nodes across languages, that would connect the semantic structure observed here to the grammatical structure of actual speech. Finally, the kinship eigenvector result deserves its own investigation. The finding that kinship terms form the densest and most mutually reinforcing cluster in the network (denser than body part vocabulary), raises the question of whether this reflects

a universal feature of social thinking or an artifact of which languages happen to be over represented in the CLICS database.

Network science is not a traditional tool in linguistics, and colexification data is not a traditional dataset for network analysis. What this project demonstrates is that the combination of the two is capable of producing findings neither would have arrived at independently: a quantitative, account of the shape of human meaning.

Acknowledgments

I used AI to assist with the formatting and syntax of the latex in this Paper. Specifically, it was used to help align figures in and out of the column spacing. Additionally it was used as a tool to refine transitional phrasing between select sections.